

Sam English
Data Science Capstone - First Outline
Professor Hardin
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Faculty and Administrative Compensation at U.S. Liberal Arts Colleges Outline

1. Introduction

Background: Administrative staffing and spending at U.S. colleges has grown faster than instructional staffing over recent decades, a trend often described as 'administrative bloat' in higher education discourse. At the same time, two major financial shocks — the 2008 recession and the COVID-19 pandemic — put sustained budget pressure on institutions, raising the question of which group, faculty or administrators, bore more of the resulting strain.

Research question: Has real compensation growth for administrative staff outpaced that of faculty, particularly non-tenure-track and visiting faculty, since the 2008 recession, and does this track a rising ratio of students-per-administrator relative to students-per-faculty?

Hypothesis: Administrative compensation and staffing capacity have grown relative to faculty compensation and staffing capacity following the 2008 and 2020 financial shocks.

2. Ethical Considerations

Aggregated institution-level averages can obscure within-rank and within-institution inequities so conclusions will be scoped to what the aggregate data can actually support.

Framing the project around "administrative vs. faculty priorities" risks an unintentionally accusatory tone toward specific institutions. Findings will be kept empirical and institution-anonymized where reasonable.

3. Data & Methods

Data: A self-built historical panel covering 19 liberal arts colleges, 2005–2024, combining IPEDS faculty salary by academic rank, IPEDS non-instructional/administrative staff salary (available 2012–2024), IPEDS enrollment and student-faculty ratio, and BLS CPI data for inflation adjustment. Regional cost-of-living adjustment (BEA Regional Price Parities) and named-administrator compensation (990 filings) are planned as next additions.

Data curation: Built using the Urban Institute's `educationdata` R package to pull IPEDS data programmatically. I have constructed resumable per-year download loops, decoded IPEDS's numeric category codes for rank, sex, and staff type and resolved inconsistent contract-length reporting across years to avoid double-counting salary totals.

Methods: A panel/fixed-effects regression (institution and year fixed effects) of real compensation on staff type (faculty rank vs. administrative), time-period indicators (pre-2008 / 2008–2019 / pandemic-and-after), and their interaction, testing whether the compensation gap widened after each shock. Paired with descriptive visualizations such as real salary trend lines by rank and staff type, and time series of students-per-faculty vs. students-per-administrator ratios.

A paired t-test will also directly compare the magnitude of real-dollar compensation change for each institution's faculty versus its administrative staff across the 2008 and 2020 shock periods, testing whether the two groups' changes differ significantly rather than relying on the regression interaction term alone.

4. Results

This section will present real salary trends by faculty rank and administrative category, staffing-ratio trends over time, and regression results testing whether compensation gaps widened after each shock.

Preliminary exploratory work suggests a more nuanced picture than hypothesized: nominal salaries rose steadily across ranks, but inflation-adjusted figures show real compensation for both faculty and administrators declining since roughly 2019, without a clear divergence between the two groups in salary terms, though the staffing-ratio data does show administrative staffing growing faster relative to enrollment than faculty staffing.

5. Discussion / Conclusion

Interpret whether the results support, complicate, or refute the original hypothesis.

Discuss implications for how institutions have prioritized administrative versus instructional capacity through financial shocks.

Limitations: administrative salary and ratio data are only available from 2012 onward, limiting direct analysis of the 2008 shock; categorizing which staff count as "administrative" involves judgment calls; regional cost-of-living adjustment and named top-administrator pay (990 data) are not yet incorporated.

Future directions: incorporate BEA Regional Price Parities, pull 990 filing data to test whether divergence is concentrated among top earners specifically, and add tenure-status breakdowns via raw IPEDS Human Resources files.

6. References / Appendix

Data sources: IPEDS (via the educationdata R package), BLS CPI, BEA Regional Price Parities (planned), NCCS/ProPublica 990 filings (planned).

Code repository: <https://github.com/sam-englishh/DS-Capstone-Project-Samuel-English>